



▲ **Figure 5.13** Flood risk in Bangladesh

Extreme events are becoming more frequent in Bangladesh. Up to 50 per cent of people living in Bangladesh's informal settlements are believed to be there due to flooding elsewhere. Dhaka, the capital, now has a population density of up to 47,500 people/km².

Tropical cyclones are becoming more intense. For example, in 2007 Cyclone Sidr brought winds of 240 km/h and killed over 3400 people. In 2009, Cyclone Aila led to the deaths of 200 people and left about 200,000 people homeless. Cyclone Amphan in 2020 was the strongest cyclone in Bangladesh's history. It led to the deaths of 128 people and caused US\$14 billion worth of damage in Bangladesh and India.

The responses to climate change

To deal with such events, Bangladesh has developed a Climate Change Strategy Action Plan (BCCSAP). This was first produced in 2009, with five adaptation plans and one for mitigation. The Nationally Determined Contributions (NDCs) of 2021 planned:

- to reduce greenhouse gas emissions by >27 million tonnes of carbon dioxide equivalent (Mt CO₂e) (6.75 per cent) using Bangladesh's own resources by the end of 2030
- to reduce greenhouse gas emissions by >60 Mt CO₂e (>15 per cent) using international support by the end of 2030
- to reduce greenhouse gas emissions by around 90 Mt CO₂e (21.5 per cent) using combined resources by the end of 2030.

Strategies and techniques to manage climate change

Adaptation projects have included the building of embankments, over 4500 cyclone shelters and 300 flood shelters, and the excavation/re-excavation of canals.

Mitigation schemes have included the planting of millions of trees, including mangrove forests. As a result, emissions are expected to have been reduced by over 2 Mt CO₂e by 2025.

Bangladesh is pursuing a low carbon future with greater emphasis on renewable energy and increased energy efficiency. Farming methods used to adapt to the changing climate include:

- early-harvest, rapid-developing varieties, for example rice that develops in 100–20 days (traditional varieties require 140–50 days)
- drought-resistant early varieties
- salt-tolerant rice varieties
- flooding-tolerant crop varieties
- stress-tolerant crop varieties.

Bangladesh joined the Kyoto Protocol in 2013 and signed the Paris Agreement in 2016.

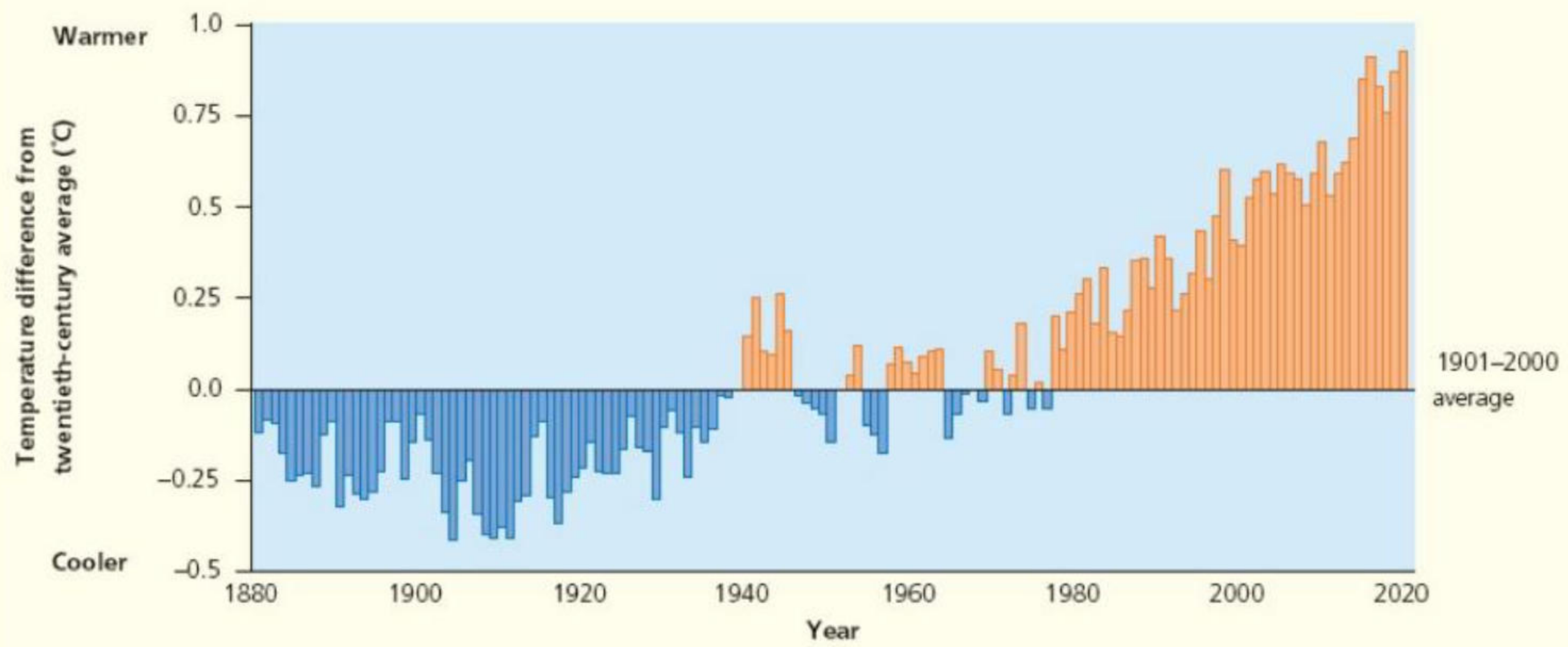
Activities

- 1 Describe the distribution and type of flood risk in Bangladesh, as shown in [Figure 5.13 \(page 120\)](#).
- 2 Distinguish between adaptation and mitigation strategies that have been used in Bangladesh.

Practice questions

- 1 [Figure 5.14](#) shows annual global surface temperature change for land and ocean, compared to the 1901–2000 average.
 - a Estimate the largest negative temperature difference from the twentieth-century average. [1]
 - b Estimate the largest positive temperature difference from the twentieth-century average. [1]
 - c Identify the decades when temperatures were lowest. [1]

- d** Identify the decade when temperatures were highest. [1]
- e** Estimate the increase in temperature between the decades with the lowest temperatures and the decade with the highest temperatures. [2]
- 2** Distinguish between the greenhouse effect and global warming. [4]
- 3 a** Define climate change mitigation and climate change adaptation. [2]
- b** Evaluate techniques of climate change mitigation and adaptation. [7]



▲ **Figure 5.14** Annual global surface temperature change for land and ocean, compared to the 1901-2000 average